# C S Vineeth

Applied Scientist · AI Researcher

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## EDUCATION

**BMS Institute of Technology and Management** Bengaluru, India

*B.E., Artificial Intelligence and Machine Learning 2019 – 2023*

## EXPERIENCE

**CBK Infotech India Pvt. Ltd.** Bengaluru, India

*Technical Consultant Apr 2023 – Present*

**Genomic & Biomedical ML**

- Constructed a reproducible ClinVar benchmark from 8.9M variants (to 637K SNVs) with principled review-status filtering, conflict resolution, and allele validation for pathogenicity research.
- Engineered 54 interpretable genomic features spanning sequence, context, and functional signals to build model-ready matrices for comparative deep-learning pathogenicity prediction.

**Medical Imaging & Computer Vision**

- Developed YOLO detection and embedding-similarity methods for privacy-preserving cross-modal dental biometrics, achieving mAP@50 0.9861 with rigorous test-set validation.
- Designed transfer-learning and knowledge-distillation frameworks (diabetic-retinopathy ordinal grading; osteoporosis from NHANES + CT/X-ray; EEG; gait), evaluating cross-dataset robustness and acquisition-shift generalization.

**Fraud Detection & Risk Intelligence**

- Investigated graph-based and transformer architectures for fraud detection on 500K+ records, benchmarking LightGBM, XGBoost, TabTransformer, and GNN variants under leakage-safe protocols to reach ROC-AUC 0.9995 / F1 0.986.
- Quantified the contribution of relational structure through graph-aggregation ablations (+2.3% accuracy) and formalized a blockchain-backed audit framework integrating model fingerprinting and input hashing for verifiable predictions.

**Cybersecurity & IoT Anomaly Detection**

- Benchmarked Random Forest, XGBoost, and LightGBM for intrusion detection on Edge-IIoTset (211K+ flows) under leakage-safe splits, using mutual information and SHAP to attribute and stress-test feature influence (AUROC > 0.999).
- Compared LSTM and Transformer autoencoders for unsupervised zero-day detection across 221,976 sequences, evaluating reconstruction-error scoring with AUROC/PR-AUC, adaptive thresholding, and calibrated open-set rejection.

**TCR Innovation** Mumbai, India

*AI Research Intern Nov 2022 – Jan 2023*

- Developed and compared computer-vision models for counting, defect detection, and segmentation (+17% over baseline).
- Researched hybrid recommendation approaches integrating collaborative, content-based, and sentiment-derived signals.
- Built reproducible preprocessing, training, and evaluation pipelines for rapid experimentation.

## SKILLS

**Languages & Data:** Python, SQL, Pandas, NumPy

**Deep Learning:** PyTorch, TensorFlow, CNNs, YOLO, LSTM, Transformers, Autoencoders, GNNs

**Machine Learning:** scikit-learn, XGBoost, LightGBM, feature selection, SHAP

**Research Methods:** Architecture benchmarking, ablation studies, cross-dataset robustness, explainability, leakage-safe validation

**Evaluation:** ROC-AUC, PR-AUC, F1, mAP, cross-validation, threshold tuning